

Proposal for Connected Minds Seed Grant

Neural Population Dynamics as a Bridge Between Computational Motor Control and Adaptive Stroke Rehabilitation

Peter Ohue¹
Gunnar Blohm^{1,2}

¹Centre for Neuroscience Studies, Queen’s University, Kingston, ON, Canada

²Department of Biomedical and Molecular Sciences, Queen’s University, Kingston, ON, Canada

Corresponding: gunnar.blohm@queensu.ca

August 2026

Project Summary (250 words)

Motor recovery after stroke is among the most pressing unmet needs in rehabilitation medicine. Despite decades of research, fewer than half of stroke survivors recover functional arm use, and existing therapies lack the mechanistic precision needed to tailor training to the individual patient. A central obstacle is that our ability to measure *why* motor adaptation succeeds or fails in a given session remains limited: conventional clinical assessments capture task outcomes but not the internal neural representations that determine whether a correction will be learned and retained.

This proposal builds directly on our recent computational work demonstrating that a reinforcement learning (RL) controller trained on adaptive obstacle-avoidance develops structured, low-dimensional neural population representations that encode perturbation context in a linearly accessible form – properties strikingly similar to those documented in biological motor cortex during reaching. We propose to extend this framework in two synergistic directions.

Aim 1. Characterise how neural population dynamics in the RL policy network predict behavioural success and failure across perturbation conditions, and test whether the representational geometry generalises to novel tasks.

Aim 2. Translate the analytical pipeline into a closed-loop virtual reality and wearable-sensor

platform for upper-limb stroke rehabilitation, using the RL agent’s latent representations as real-time biomarkers to personalise training difficulty and provide quantitative outcome measures beyond conventional clinical scales.

The project directly advances the Connected Minds priority of human-centred AI for health, generating tools and insights applicable to any motor disorder in which adaptation under uncertainty is impaired.

Background and Rationale

The Clinical Problem

Stroke is a leading cause of persistent adult disability in Canada, with approximately 62,000 new strokes occurring each year and upper-limb motor impairment affecting the majority of survivors [1]. Recovery is highly heterogeneous: some patients regain near-normal function while others plateau at severe impairment, and current clinical tools cannot reliably predict which trajectory a given patient will follow [2]. High-dose, task-oriented practice is the most evidence-supported rehabilitation strategy [3], but translating this principle into individualized, adaptive protocols – ones that automatically adjust challenge level to the patient’s changing capacity – remains an open engineering and scientific problem.

The Scientific Gap

The neural mechanisms underlying motor re-learning after stroke are poorly understood at the population level. Research in healthy adults has established that motor cortex activity during reaching organises onto low-dimensional neural manifolds whose geometry changes systematically with learning [4, 5, 6]. However, it is unknown whether similar manifold reorganisation occurs during stroke recovery, whether the rate of that reorganisation predicts clinical outcomes, or how external perturbation schedules should be designed to accelerate it.

Our recent RL-based work provides a computational bridge. We showed that a PPO controller trained purely by reward on an obstacle-avoidance task spontaneously develops hidden-layer representations with the same organisational properties – low dimensionality, phase-aligned clustering, graded kinematic encoding, linear decodability – observed in biological motor cortex [7]. Critically, the latent states of successfully correcting trials are geometrically distinct from those of failure trials, suggesting that population geometry is a principled indicator of adaptive capacity. Translating this observation into a clinical tool requires: (i) extending the RL framework to richer, human-relevant environments; and (ii) acquiring a corresponding dataset from human stroke survivors performing analogous reaching tasks.

Connected Minds Alignment

This proposal aligns with Connected Minds’ focus on digital technologies that augment human cognitive and physical capability. The planned closed-loop VR-RL platform is an instance of human-centred AI: the RL agent models the patient’s evolving motor policy in real time, and its latent representations drive an adaptive training protocol that is responsive to the individual’s neural and kinematic state rather than following a fixed schedule. The project also advances equity of access by targeting a platform that is deployable in home settings, reducing dependence on specialized rehabilitation centres.

Research Objectives and Hypotheses

Overall objective: To establish low-dimensional latent neural population dynamics, derived from both computational RL models and from kinematic/EMG surrogates in human participants, as quantitative biomarkers of adaptive motor learning and as drivers of individualized rehabilitation protocols.

Hypothesis 1. The representational geometry of a trained RL policy’s hidden-layer activity will generalise across task variants: the same low-dimensional manifold structure and phase-aligned clustering identified in the obstacle-avoidance task will emerge in a richer perturbation environment with time-varying force fields and dynamic obstacles.

Hypothesis 2. In stroke survivors, the rate of manifold reorganisation during a standardised perturbation-based reaching protocol – quantified by the change in population variance explained by the first two PCs across sessions – will correlate significantly with gains on the Fugl-Meyer Upper Extremity (FMA-UE) scale at six-week follow-up.

Hypothesis 3. A closed-loop training protocol in which task difficulty is modulated by the RL agent’s latent state geometry (specifically, by the Euclidean distance of the current trial’s latent trajectory from the learned “success manifold”) will produce larger FMA-UE gains over six weeks than a matched fixed-difficulty protocol, in a randomised single-site pilot trial.

Methodology

Phase 1: Extended Computational Framework (Months 1–8)

Environment extension. We will extend the current two-dimensional point-mass environment to include: (i) a three-degree-of-freedom planar arm with muscle-like actuation and joint limits; (ii) time-varying force fields whose magnitude and direction are drawn from the same distribution as the clinical perturbation protocol; and (iii) dynamic obstacles that require online re-planning. The RL controller will be trained using PPO with the same

hyperparameter schedule validated in the baseline study, and the neural population analysis pipeline (PCA, K -means clustering, linear decoding) will be applied to the hidden-layer activity of the converged policy.

Multi-seed validation. A minimum of ten independent training runs will be conducted to characterise the variability in representational structure across seeds. Features that are stable across seeds (low coefficient of variation) will be designated as candidate biomarkers for translation to the clinical phase.

Recurrent architecture comparison. A long short-term memory (LSTM) variant of the policy network will be trained under identical conditions. Comparing the manifold geometry between feedforward MLP and LSTM policies will reveal whether temporal memory substantially changes the representational organisation, informing the choice of architecture for the closed-loop platform.

Phase 2: Human Participant Study (Months 6–18)

Participants. We will recruit $n = 20$ healthy adults and $n = 30$ chronic stroke survivors (FMA-UE 20–55, ≥ 6 months post-stroke, sufficient cognitive function for VR interaction). The sample size for the stroke group is powered to detect a correlation of $r \geq 0.50$ between manifold reorganisation rate and FMA-UE change ($\alpha = 0.05$, $1 - \beta = 0.80$, two-tailed).

Instrumentation. Participants will perform a standardised perturbation-based reaching task in a custom VR environment rendered on a head-mounted display (Meta Quest Pro or equivalent). Reaching kinematics will be captured at 100 Hz using wrist-worn IMUs (XSENS DOT) and processed to extract the same eight-dimensional state vector used in the RL environment (goal-relative position, obstacle-relative position, velocity). Surface EMG will be recorded from six upper-limb muscles to provide a neural surrogate for population analysis.

Perturbation protocol. Seven perturbation conditions mirroring the computational study (P0, L1–L3, R1–R3) will be applied via a lightweight cable-driven exoskeleton (hand-exo or equivalent) or through VR-mediated haptic redirection. Three sessions per week for six weeks will be administered, with each session comprising 100 reaching trials.

Population analysis. PCA and linear decoding will be applied to the EMG/kinematic population vector at each timestep, paralleling the RL analysis pipeline. The primary outcome for Hypothesis 2 is the session-by-session change in % variance explained by PC1+PC2 of the EMG population, correlated with FMA-UE change scores.

Phase 3: Closed-Loop Pilot Trial (Months 14–24)

Adaptive protocol design. The difficulty-adaptation rule will use the latent state geometry from the RL model as a reference manifold. On each trial, the deviation of the participant’s

kinematic state trajectory from the RL “success manifold” (projected into the common PCA space) will be computed online. If the deviation exceeds a threshold calibrated from the Phase 2 data, the perturbation magnitude will be reduced by one step; if it falls below a second threshold, the perturbation magnitude will be increased. This rule mimics the bandwidth feedback used in motor learning research and is consistent with the principles of optimal challenge point theory.

Randomised pilot. $n = 20$ stroke survivors will be randomised (1:1) to adaptive vs. fixed-difficulty six-week protocols. Primary outcome: FMA-UE change at six weeks. Secondary outcomes: kinematic smoothness (SPARC), manifold reorganisation rate, number of sessions required to reach each perturbation level, patient-reported experience (NASA-TLX).

Statistical analysis. The primary analysis is a mixed-model repeated-measures ANOVA with group and session as factors. Effect sizes (Cohen’s d) will be reported with 95% confidence intervals. This is a feasibility and signal-detection trial; results will be used to power a future definitive RCT.

Team and Expertise

Gunnar Blohm (PI) is a Professor and Canada Research Chair at the Centre for Neuroscience Studies, Queen’s University, with expertise in computational motor neuroscience, sensorimotor transformations, and neural population modelling. His laboratory has a track record of developing and validating population- level analyses of motor cortex and brainstem circuits.

Peter Ohue (Graduate Trainee / Co-I) is a doctoral candidate in the Blohm laboratory with expertise in reinforcement learning, neural population analysis, and computational neuroscience. He is the primary developer of the RL-based obstacle-avoidance framework on which this proposal is based, and will lead the computational and experimental phases.

Collaborators (to be confirmed): A rehabilitation medicine specialist and a certified occupational therapist will provide clinical expertise for the stroke participant protocol and outcome assessment. A biomedical engineer with wearable-sensor expertise will support the IMU and EMG data acquisition pipeline.

Expected Outcomes and Impact

1. An extended RL framework with validated multi-seed representational stability, providing a robust computational foundation for clinical translation.
2. A normative dataset of kinematic manifold dynamics in healthy adults and stroke survivors during standardised perturbation-based reaching, establishing reference values for

the proposed biomarkers.

3. Proof-of-concept evidence for the validity of RL latent geometry as a predictor of clinical rehabilitation outcomes.
4. A pilot RCT comparing adaptive vs. fixed-difficulty VR-RL rehabilitation, with effect size estimates to power a future definitive trial.
5. Open-source code and data deposited on Zenodo and GitHub, enabling independent replication and extension.

The broader impact extends beyond stroke: the framework is applicable to any motor disorder in which adaptation under perturbation is impaired, including Parkinson’s disease, cerebellar ataxia, and spinal cord injury. The home-deployable VR-wearable platform addresses the access gap between specialised rehabilitation centres and community care, directly advancing Connected Minds’ mandate of equitable, technology-enabled human flourishing.

Timeline

Period	Milestones
Months 1–4	Environment extension (3-DOF arm, force fields, dynamic obstacles); multi-seed training; LSTM comparison
Months 3–6	Ethics approval (human study); wearable instrumentation setup; pilot feasibility in healthy adults ($n = 5$)
Months 6–12	Healthy adult dataset ($n = 20$); manifold analysis pipeline validation against RL reference
Months 10–18	Stroke survivor observational study ($n = 30$); Hypothesis 2 analysis
Months 14–24	Adaptive-protocol pilot RCT ($n = 20$ stroke survivors); Hypothesis 3 analysis
Months 22–24	Manuscript preparation; data deposition; dissemination at FENS, SFN, and Canadian neuroscience meetings

Budget Justification

Note: Amounts are indicative for a Connected Minds Seed Grant (typically \$50,000–\$100,000 over 1–2 years). Adjust to the specific call ceiling and duration.

Category	Amount (CAD)	Justification
Graduate student stipend (1 FTE, 24 months)	\$60,000	Peter Ohue (doctoral trainee); standard Queen's University stipend (\$30,000/year)
Research assistant (part-time, 12 months)	\$18,000	Experimental support for human participant sessions
Wearable sensors (XSSENS DOT $\times 4$)	\$8,000	Kinematic capture for human participant protocol
VR hardware (Meta Quest Pro or equivalent)	\$4,000	Head-mounted display for VR reaching task
EMG system (Delsys Trigno $\times 6$ ch.)	\$6,000	Neuromuscular signal acquisition
Computing (cloud GPU credits)	\$3,000	Multi-seed RL training; real-time latent state computation
Participant compensation (\$20/h, 50 h total)	\$4,000	Healthy adults and stroke survivors
Open-access publication fees (2 papers)	\$6,000	PLOS Computational Biology or equivalent
Conference travel (2 conferences)	\$4,000	SFN, Canadian Association for Neuroscience
Indirect costs (20%)	\$22,600	University overhead rate
Total	\$135,600	

For a \$50,000 ceiling: reduce to 12 months, one conference, no EMG system; request the EMG as in-kind from the rehabilitation collaborator.

Data Management and Open Science

All experimental data will be de-identified and deposited on the Queen's University institutional repository (QSPACE) and on Zenodo with a persistent DOI within six months of collection. Analysis code and trained RL models will be maintained on GitHub (already public: github.com/OhuePeter/NeuroRL-ObstacleAvoidance-v1.0) under the MIT licence. Manuscripts will be submitted to open-access journals or posted as preprints on bioRxiv

concurrent with journal submission.

Ethics and Risk

Human participant research will be conducted under Queen’s University Health Sciences and Affiliated Teaching Hospitals Research Ethics Board (HSREB) approval, which will be sought in Months 3–5. The perturbation protocol uses brief, low-force lateral impulses comparable in magnitude to those used routinely in motor learning research and poses negligible physical risk. VR exposure will be limited to 30-minute sessions to minimise cybersickness. Stroke participants will be screened to exclude severe spasticity, uncontrolled epilepsy, and visual impairments.

Knowledge Translation

Short-term (<2 years): peer-reviewed publications in computational neuroscience and rehabilitation journals; open-source code and data release; conference presentations.

Medium-term (2–5 years): submission of a full-scale CIHR Project Grant or NSERC Discovery Grant building on the Seed results; regulatory-pathway consultation for clinical device classification of the closed-loop platform.

Long-term (>5 years): commercialisation pathway in partnership with Queen’s Partnerships and Innovation (QPI) for a CE/FDA-clear home rehabilitation device; integration with publicly funded rehabilitation programmes in Ontario.

References

- [1] Heart & Stroke Foundation of Canada. Stroke statistics. <https://www.heartandstroke.ca/stroke/what-is-stroke>, 2023.
- [2] Gert Kwakkel, Natasha A. Lannin, Karen Borschmann, Coralie English, Myzoon Ali, Leonid Churilov, Gustavo Saposnik, Carolee Winstein, Erwin E. H. van Wegen, Steven L. Wolf, John W. Krakauer, and Julie Bernhardt. Standardized measurement of sensorimotor recovery in stroke trials: Consensus-based core recommendations from the stroke recovery and rehabilitation roundtable. *International Journal of Stroke*, 12(5):451–461, 2017.
- [3] B. French, L. H. Thomas, J. Coupe, N. E. McMahon, L. Connell, J. Harrison, C. J. Sutton, S. Tishkovskaya, and C. L. Watkins. Repetitive task training for improving functional ability after stroke. *Cochrane Database of Systematic Reviews*, 11:CD006073, 2016.

- [4] Mark M. Churchland, John P. Cunningham, Matthew T. Kaufman, Justin D. Foster, Paul Nuyujukian, Stephen I. Ryu, and Krishna V. Shenoy. Neural population dynamics during reaching. *Nature*, 487(7405):51–56, 2012.
- [5] Patrick T. Sadtler, Kristin M. Quick, Matthew D. Golub, Steven M. Chase, Stephen I. Ryu, Elizabeth C. Tyler-Kabara, Byron M. Yu, and Aaron P. Batista. Neural constraints on learning. *Nature*, 512(7515):423–426, 2014.
- [6] Emily R. Oby, Matthew D. Golub, Joram A. Hennig, Alan D. Degenhart, Elizabeth C. Tyler-Kabara, Byron M. Yu, Steven M. Chase, and Aaron P. Batista. New neural activity patterns emerge with long-term learning. *Proceedings of the National Academy of Sciences*, 116(30):15210–15215, 2019.
- [7] Peter Ohue, Emily Oby, and Gunnar Blohm. Neural population dynamics reveal internal representations underlying adaptive obstacle avoidance in reinforcement learning. *bioRxiv preprint*, 2026.